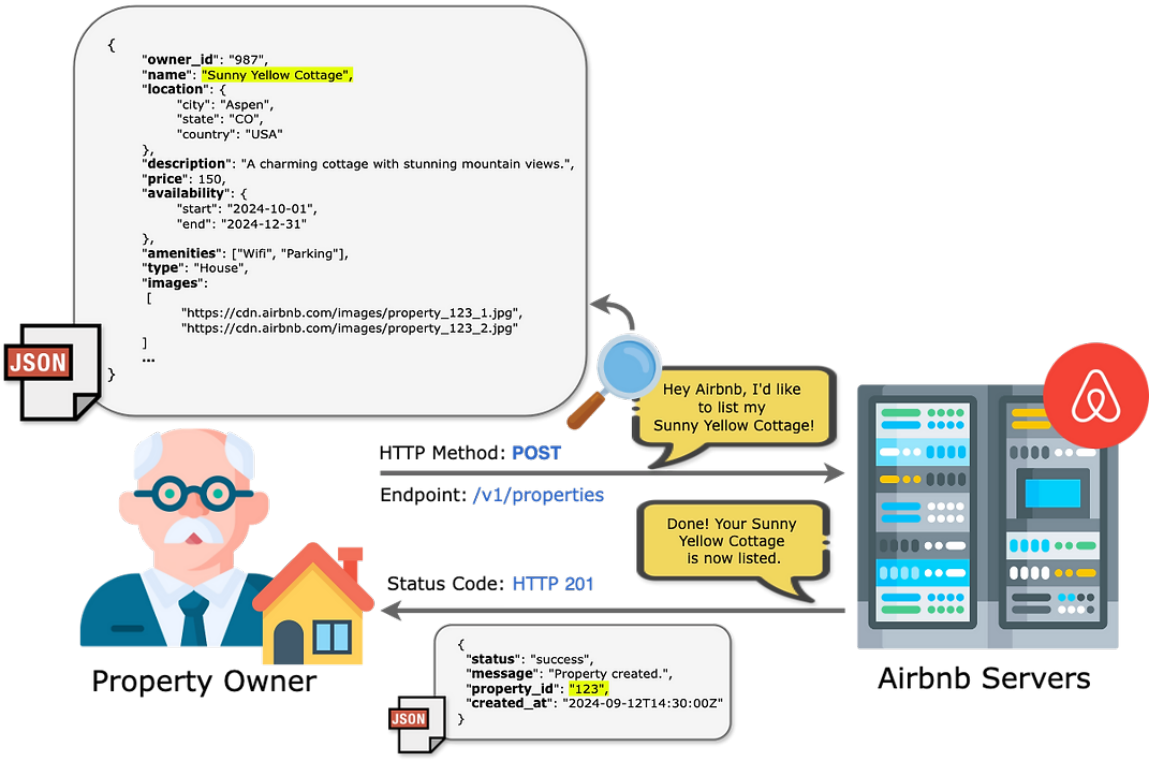
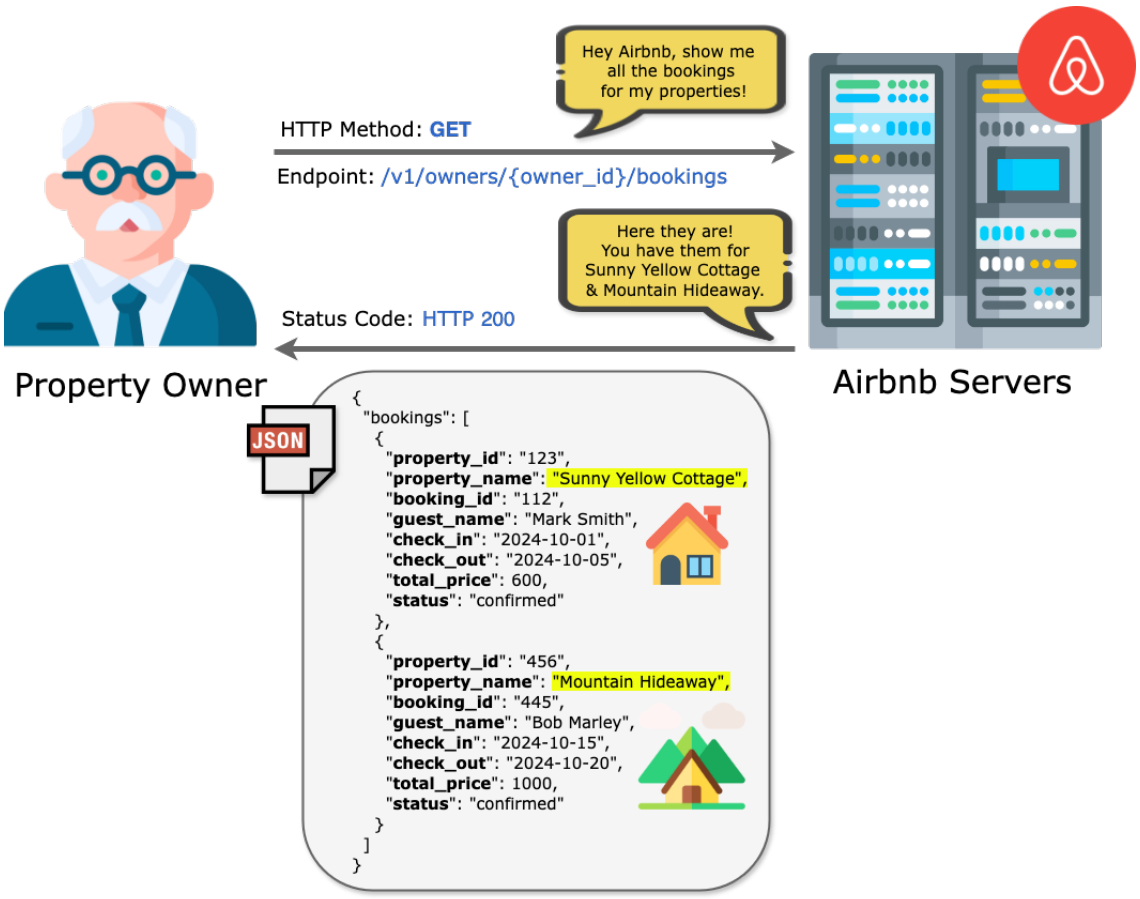
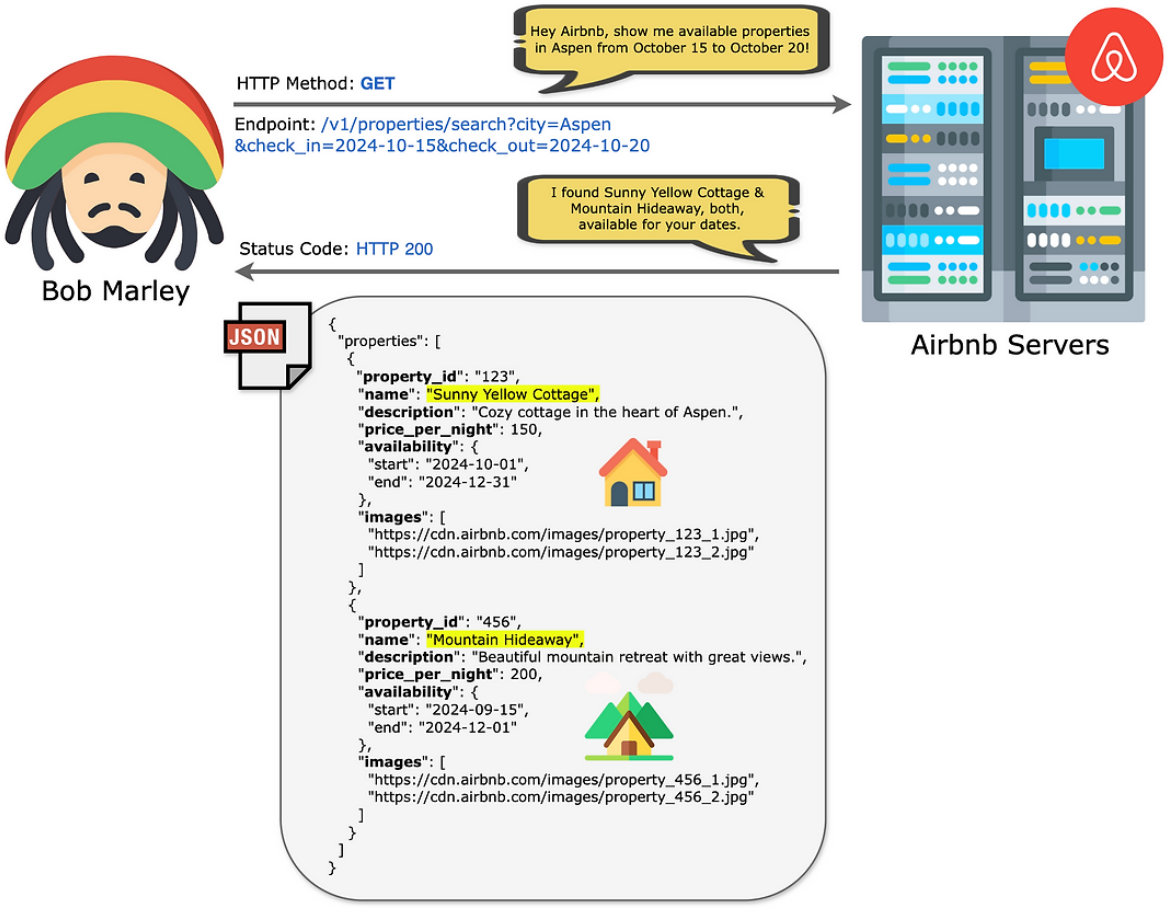
API Design-Add Property (Property Owner)



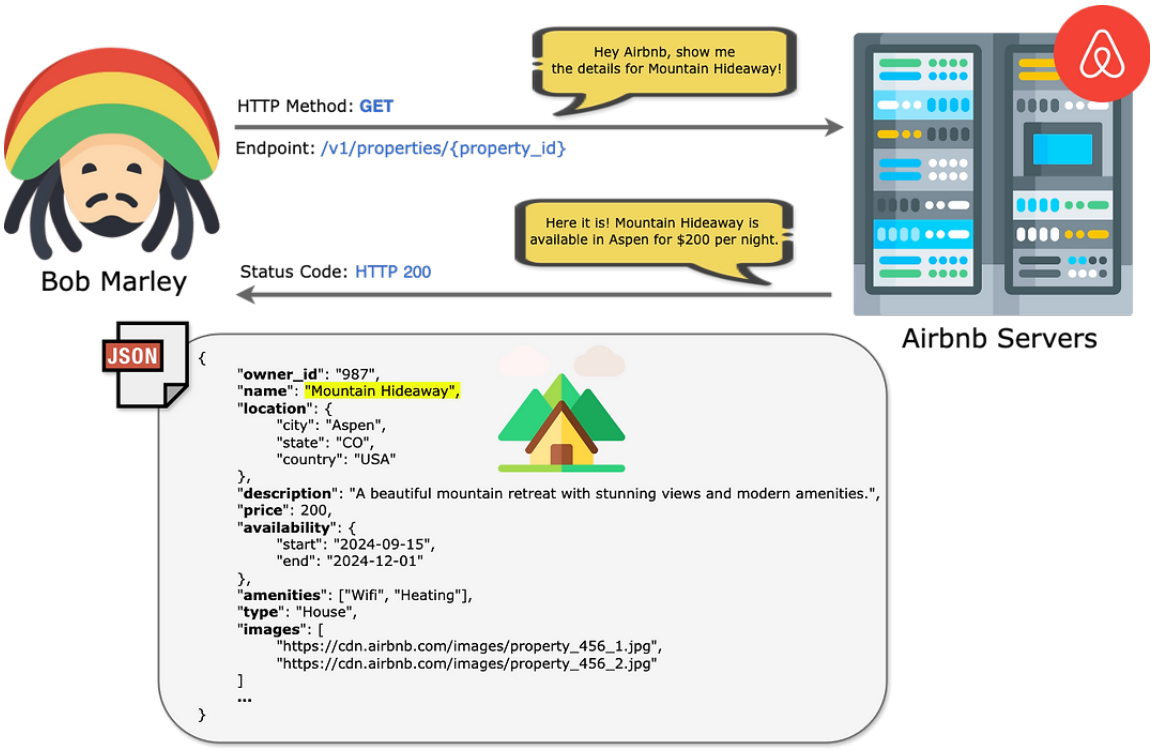
API Design-View Bookings (Property Owner)



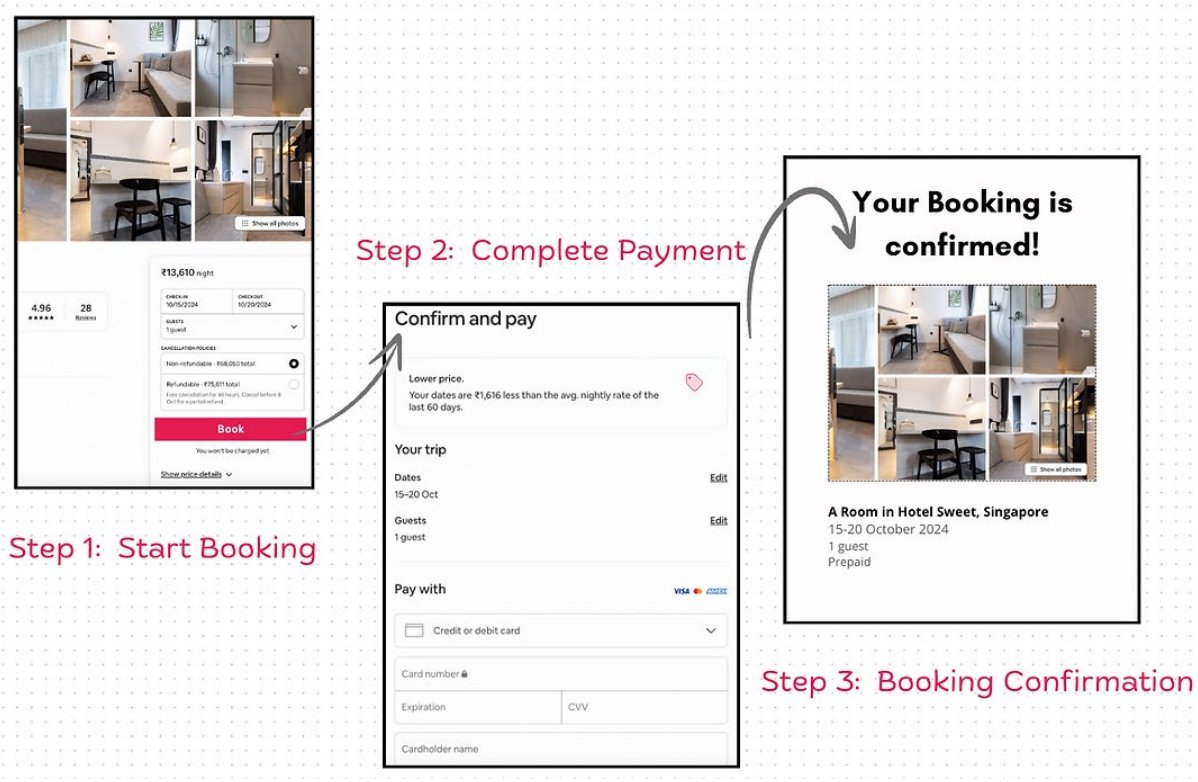
API Design-Search Properties (Guest)

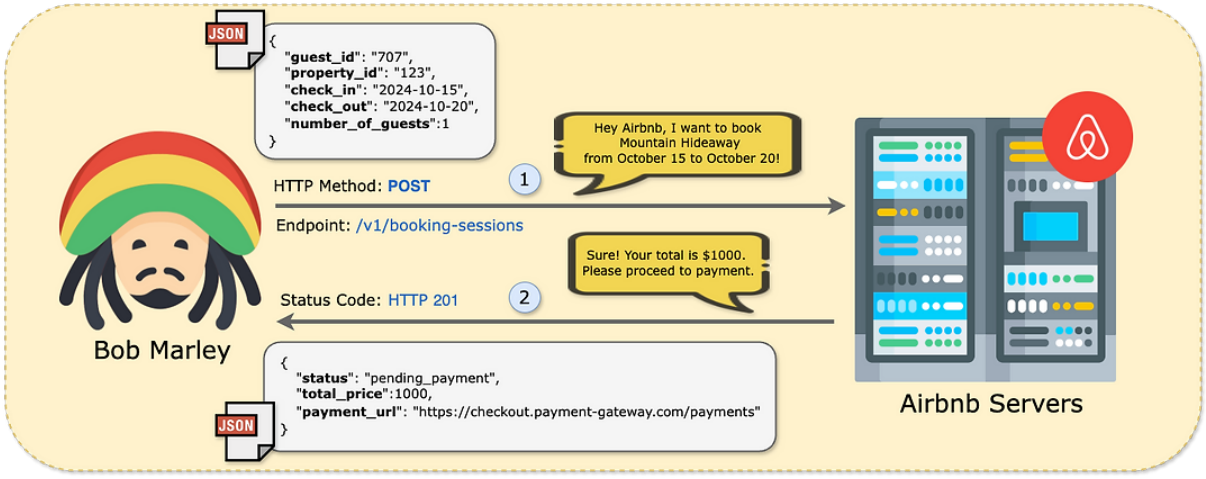


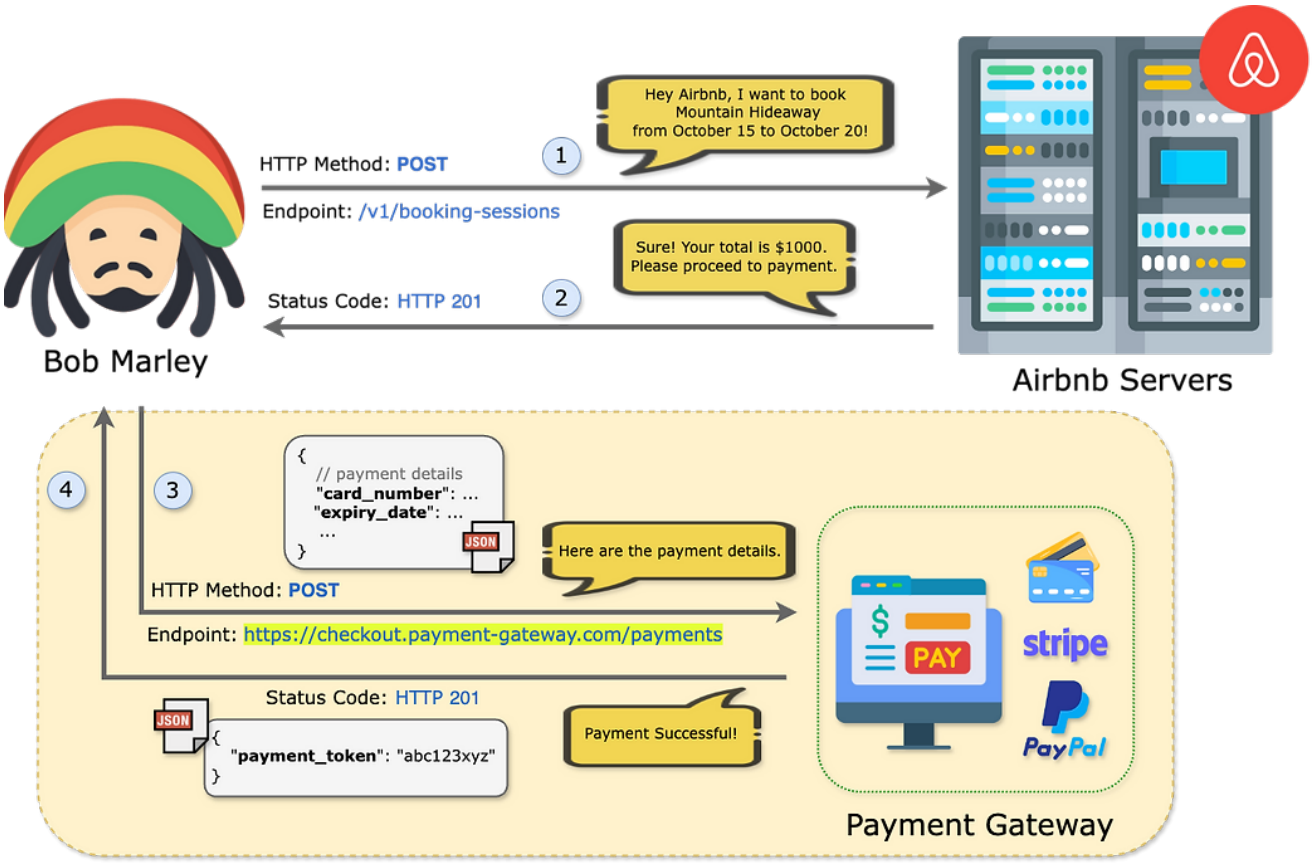
API Design-View Property (Guest)

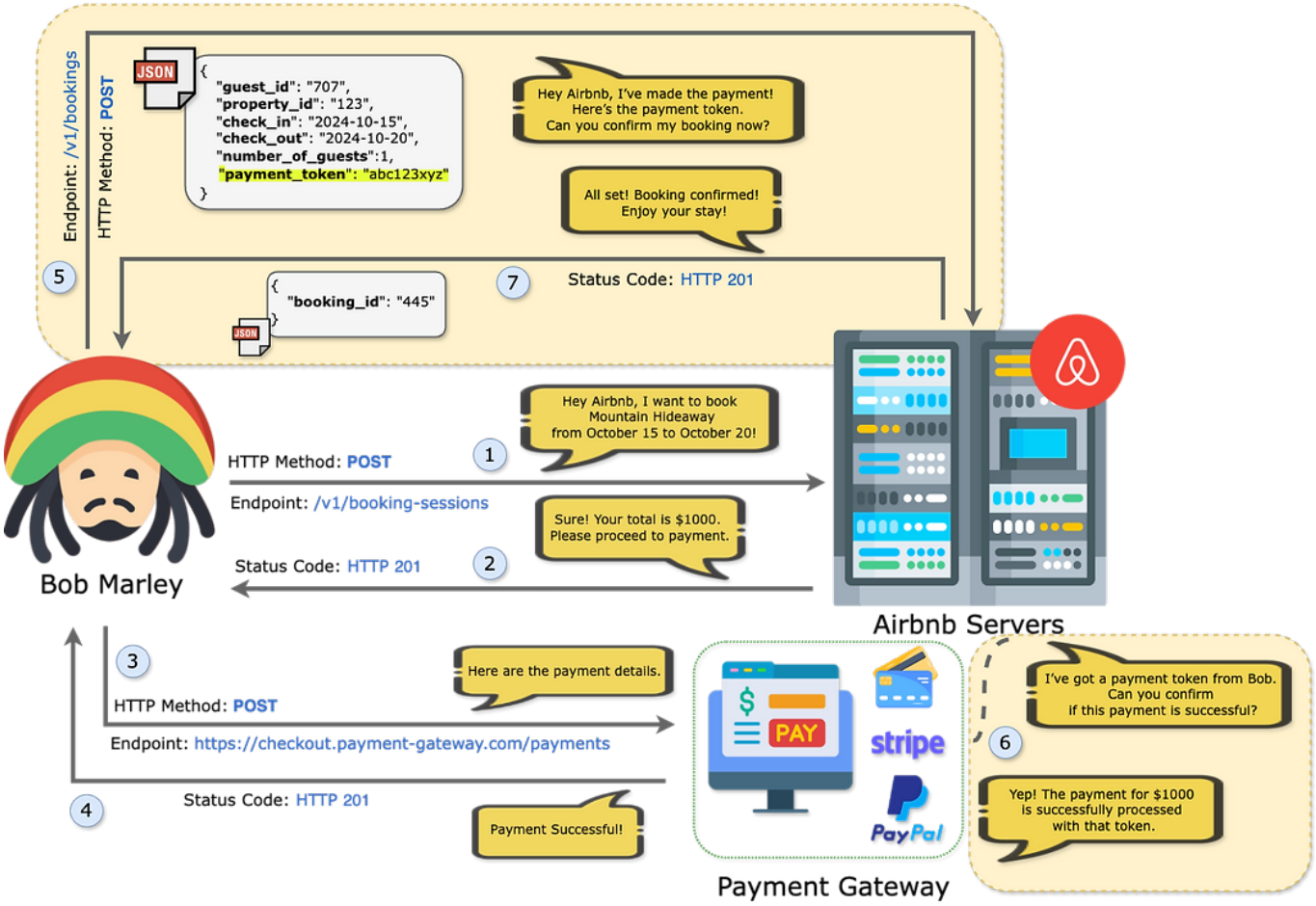


API Design-Book Property (Guest)

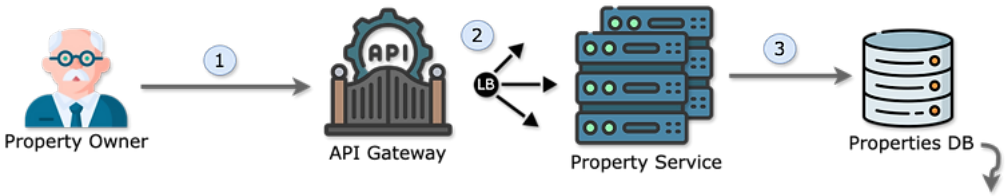




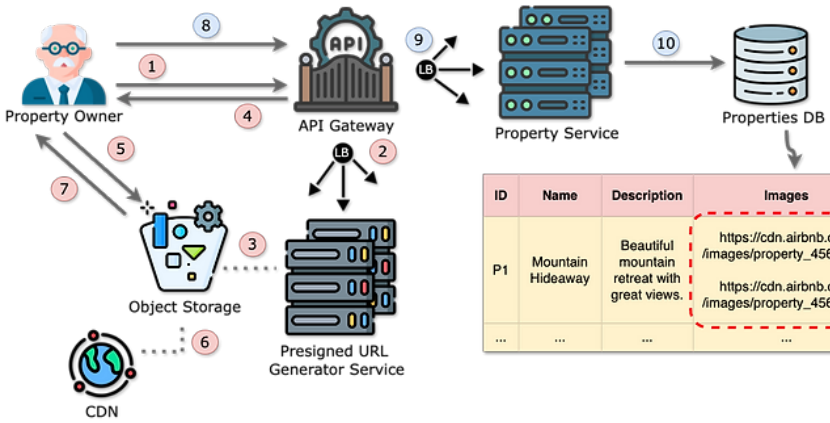

Bob Marley
HTTP Method: POST
Endpoint: /v1/booking-sessions
1
Hey Airbnb, I want to book Mountain Hideaway from October 15 to October 20!
Status Code: HTTP 201
2
Sure! Your total is $1000. Please proceed to payment.
Airbnb Servers
3
4
{
// payment details
"card_number": ...
"expiry_date": ...
...
}
JSON
Here are the payment details.
HTTP Method: POST
Endpoint: https://checkout.payment-gateway.com/payments
Status Code: HTTP 201
JSON
{
"payment_token": "abc123xyz"
}
Payment Successful!
PAY
stripe
PayPal
Payment Gateway


Endpoint: /v1/bookings
HTTP Method: POST
JSON
{
"guest_id": "707",
"property_id": "123",
"check_in": "2024-10-15",
"check_out": "2024-10-20",
"number_of_guests":1,
"payment_token": "abc123xyz"
}
Hey Airbnb, I've made the payment! Here's the payment token. Can you confirm my booking now?
All set! Booking confirmed! Enjoy your stay!
5
7
Status Code: HTTP 201
{
"booking_id": "445"
}
JSON
Bob Marley
HTTP Method: POST
1
Hey Airbnb, I want to book Mountain Hideaway from October 15 to October 20!
Endpoint: /v1/booking-sessions
Status Code: HTTP 201
2
Sure! Your total is $1000. Please proceed to payment.
Airbnb Servers
3
Here are the payment details.
HTTP Method: POST
Endpoint: https://checkout.payment-gateway.com/payments
4
Status Code: HTTP 201
Payment Successful!
PAY
stripe
PayPal
Payment Gateway
I've got a payment token from Bob. Can you confirm if this payment is successful?
6
Yep! The payment for $1000 is successfully processed with that token.

High Level Design-Add/Update property

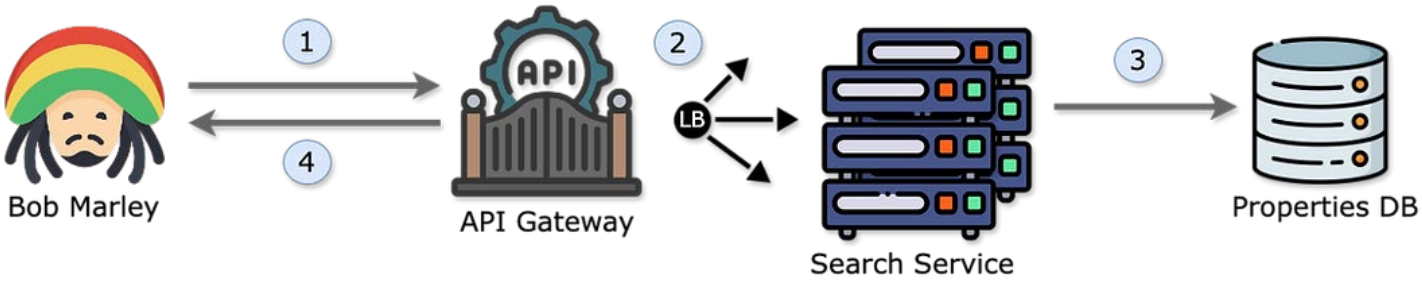


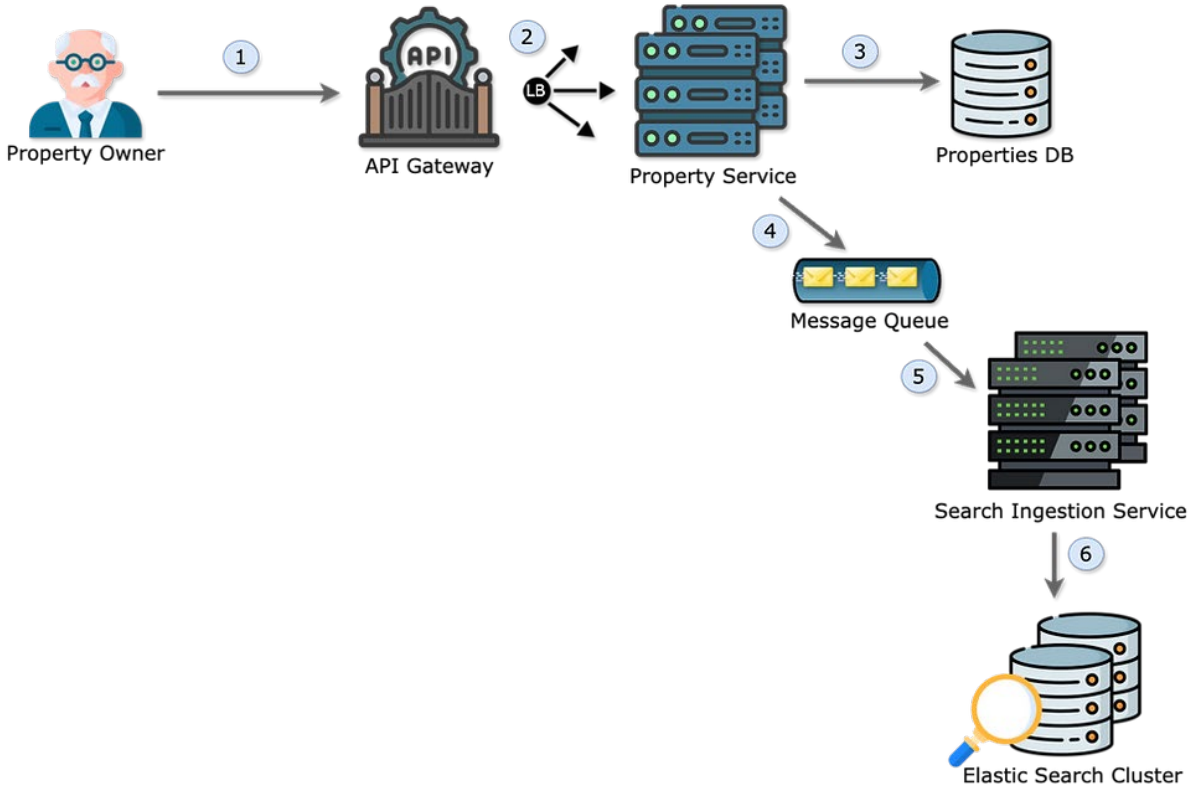
ID	Name	Description	Images	Price	Available Start Dates	Available End Dates	OwnerID	City	State	Country	...
P1	Mountain Hideaway	Beautiful mountain retreat with great views.	N/A	200	2024-09-15	2024-12-01	987	Aspen	CO	USA	...
...	...	...	...	...	...	...	...	...	...	...	...



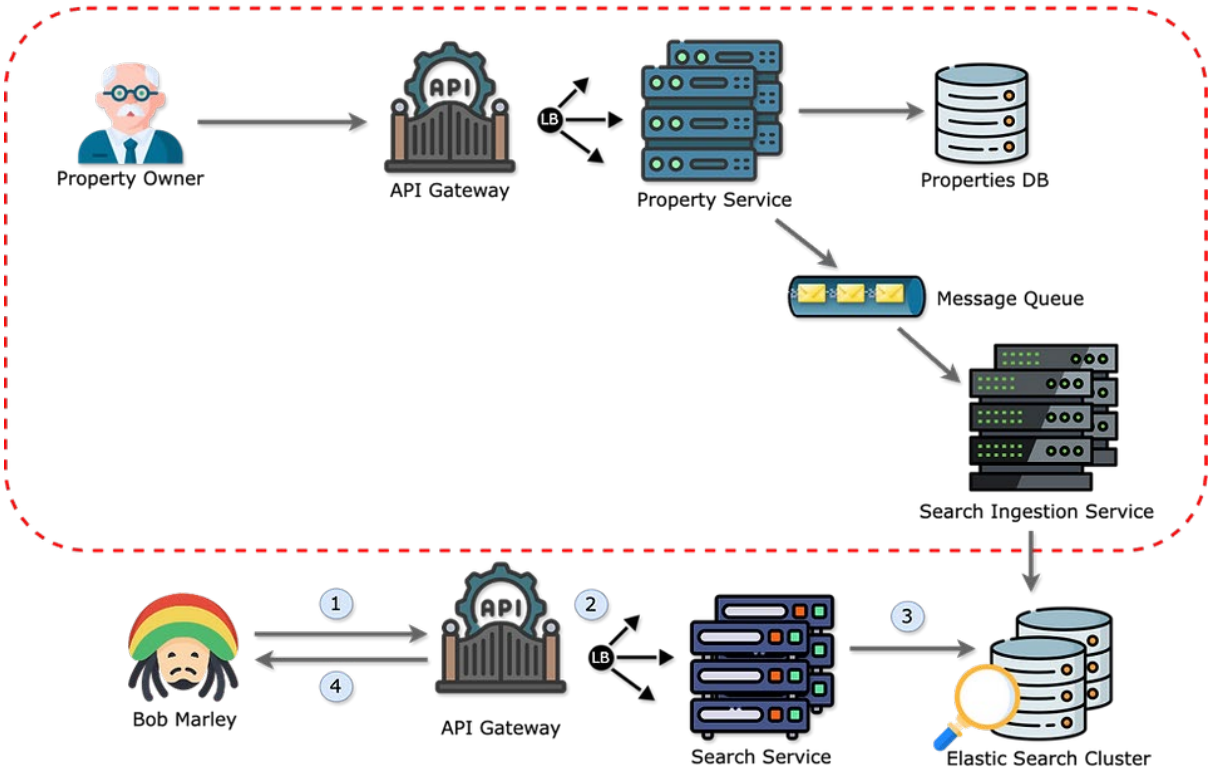
ID	Name	Description	Images	Price	Available Start Dates	Available End Dates	OwnerID	City	State	Country	...
P1	Mountain Hideaway	Beautiful mountain retreat with great views.	https://cdn.airbnb.com /images/property_456_1.jpg https://cdn.airbnb.com /images/property_456_2.jpg	200	2024-09-15	2024-12-01	987	Aspen	CO	USA	...
...	...	...	...	...	...	...	...	...	...	...	...

High Level Design- Search Property (Guest)

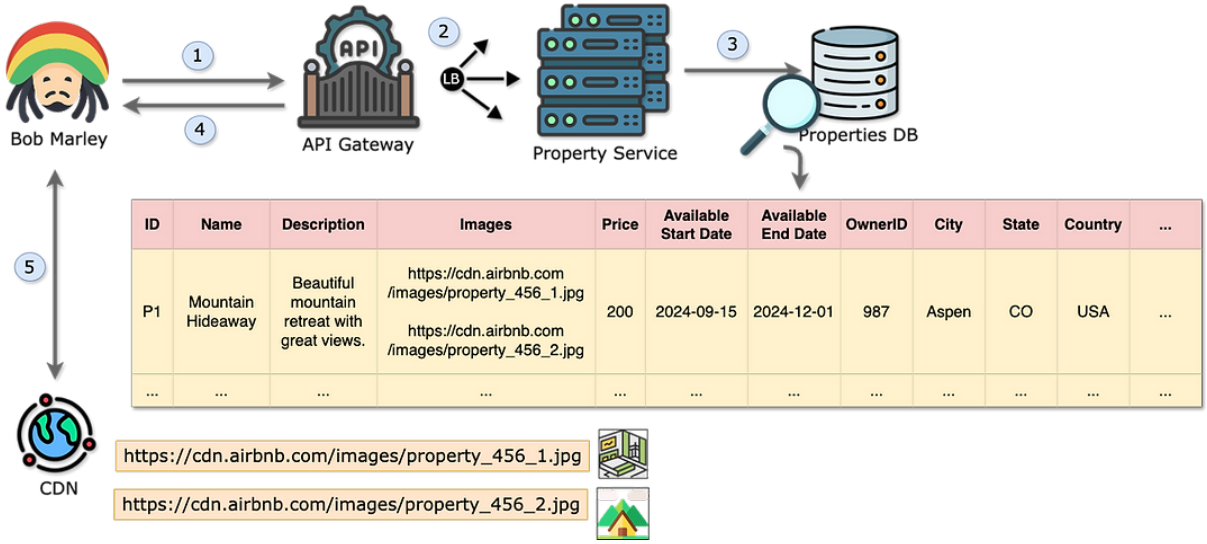




Search Data 'Pre'-Processing



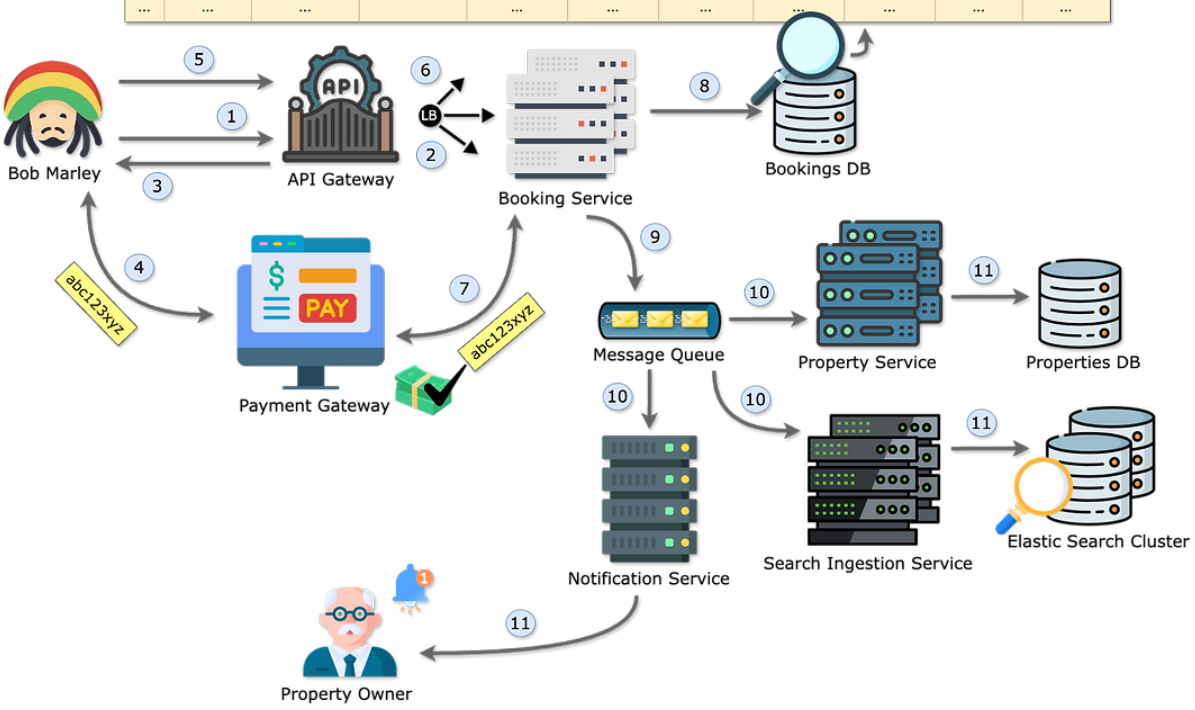
High Level Design-View Property (Guest)



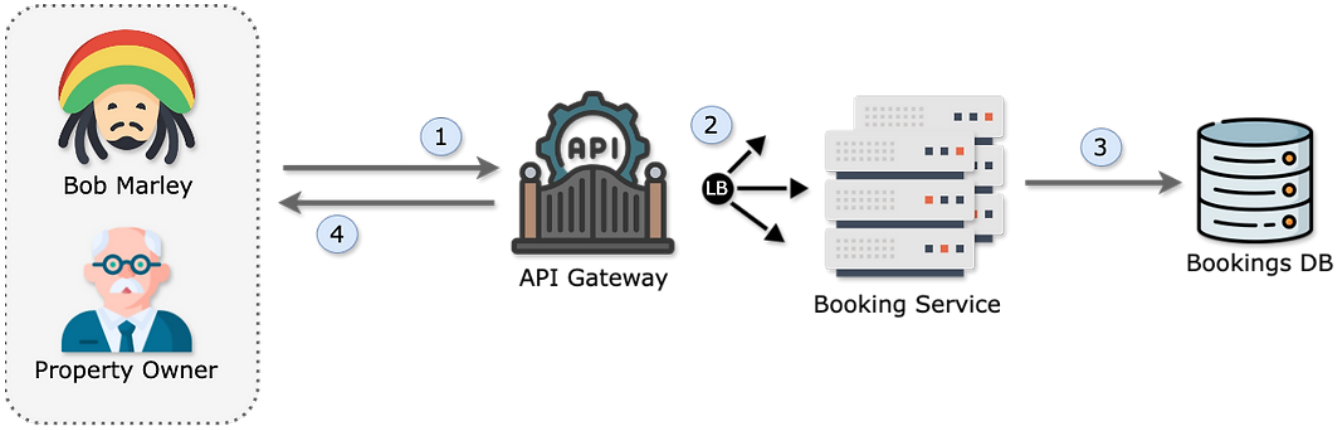
ID	Name	Description	Images	Price	Available Start Date	Available End Date	OwnerID	City	State	Country	...
P1	Mountain Hideaway	Beautiful mountain retreat with great views.	https://cdn.airbnb.com /images/property_456_1.jpg https://cdn.airbnb.com /images/property_456_2.jpg	200	2024-09-15	2024-12-01	987	Aspen	CO	USA	...
...	...	...	...	...	...	...	...	...	...	...	...

High Level Design-Book Property (Guest)

ID	Property ID	Guest ID	Guest Name	Owner ID	Check In	Check Out	Total Price	Payment Token	Booking Status	...
B1	P1	707	Bob Marley	987	2024-10-15	2024-10-20	1000	abc123xyz	Confirmed	...
...	...	...		...	...	...	...	...	...	...



High Level Design-View Bookings (Property Owner/Guest)



Properties DB Schema

{

PropertyID: A unique ID for each property listing.
Name: The name of the property.
Description: A brief description of the property.
Images: URLs to the property images.
Price: Price per night.
AvailableStartDates: List of start dates when the property is available.
AvailableEndDates: List of end dates when the property is no longer available.
OwnerID: ID of the property owner.
City: City where the property is located.
State: State where the property is located.
Country: Country where the property is located.
Amenities: ...
...

}

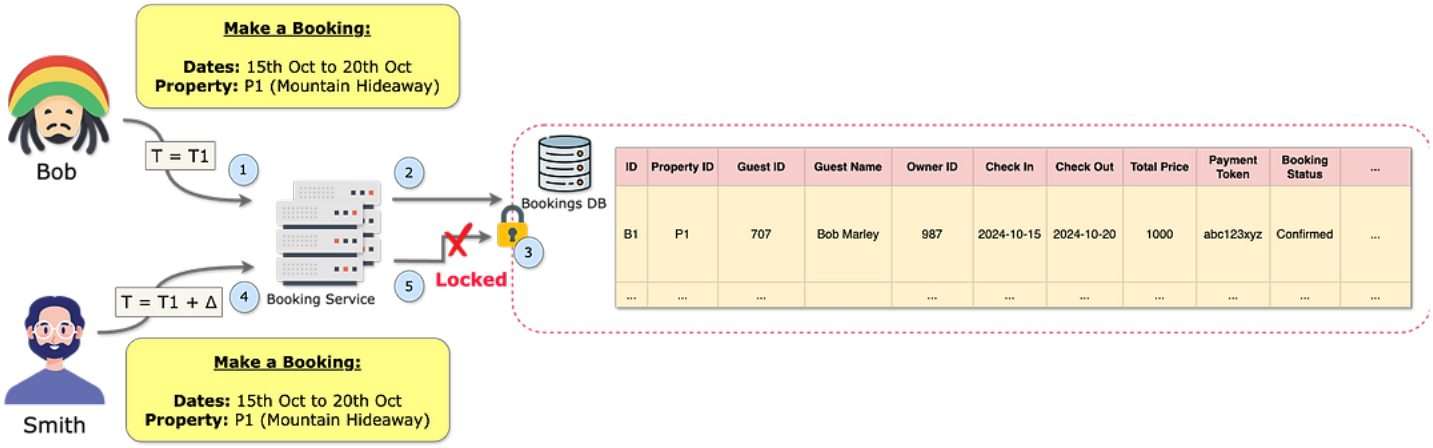
Bookings DB Schema

{

BookingID: A unique identifier for each group.
PropertyID: The ID of the property being booked.
GuestID: The ID of the guest making the booking.
GuestName: Name of the guest.
OwnerID: The ID of the property owner.
CheckInDate: Date when the guest will check in.
CheckOutDate: Date when the guest will check out.
TotalPrice: Total price for the stay.
PaymentToken: Token received after successful payment.
BookingStatus: Status of the booking (e.g., "Confirmed", "Completed", "Cancelled")
...

}

Handling Concurrent Bookings



ID	Property ID	Guest ID	Guest Name	Owner ID	Check In	Check Out	Total Price	Payment Token	Booking Status	...
B1	P1	707	Bob Marley	987	2024-10-15	2024-10-20	1000	abc123xyz	Confirmed	...
...	...	...		...	...	...	...	...	...	...